THE BLACK BOX

Transparency, opacity, and the limits of explanation in AI

A VADEMECUM



Input and output are visible. The reasoning that connects them may not be.

What is a black box?

A system where you see the input and the output — but not the reasoning that connects them.



The box worked — but why did it say that? With a deep neural network, the answer is spread across millions of numerical weights, with no single human-readable rule to point to.

It is not that the answer is hidden from you — it may not exist in any form a human can read.

Two very different kinds of opacity

The word "black box" hides two problems with completely different ethical weight.

1 — Opacity by choice

Secrecy

The knowledge exists — someone is choosing not to share it.

- Trade secret, proprietary model, "we can't disclose that."
- Someone could open the box; they choose not to.
- Solvable by decision, regulation, or force.

2 — Opacity by nature

Intrinsic — the hard one

Even the builders can't fully explain a given decision.

- Not secrecy — there is no readable mechanism inside to reveal.
- Nobody can open this box.
- You cannot regulate away the fact that a network doesn't reason in sentences.

Why it matters: opacity leaks accountability

Responsibility follows knowledge. The black box can destroy the knowledge — for everyone.

The doctor

can't interrogate a reason that isn't there.

The model

can't be morally accountable for anything.

The builders

"we didn't choose this output — the trained system produced it."

The black box is the technical engine of the responsibility gap. When no one can see the reasoning, accountability falls into the space between the actors.

The trade-off nobody likes to state

The most accurate models are often the least interpretable — and vice versa.

Transparent model

- You fully understand its logic.
- Decision trees, linear models — readable rules.
- But often less accurate.
- Cost: it misses more sick patients.

Opaque model

- Catches more illness — higher accuracy.
- The complexity that makes it opaque is what makes it good.
- But you cannot explain or fully contest its decisions.

"Always choose interpretability" can mean choosing to let more people die for the sake of explainability. That is a real moral cost — not a free win.

Explainable AI — a partial escape

Techniques (XAI) that give approximate, after-the-fact explanations for a black box's decisions.

What it does

- Offers a human-readable story: "this diagnosis was driven mostly by these three factors."
- Genuinely useful for spotting gross errors and building trust.

The catch

- These are reconstructions, not the real reasoning.
- A plausible story that is subtly wrong can be worse than honest silence.
- It manufactures false confidence that we understand the box.

An explanation you trust but that is subtly wrong is its own kind of danger.

Vademecum — the essentials

Definition

A black box: inputs and outputs visible, the reasoning between them not.

Two opacities

By choice (secrecy — solvable) vs by nature (intrinsic — nobody can open it).

The trade-off

Accuracy vs interpretability is a genuine dilemma, not a free choice.

The gap

Opacity is the engine of the responsibility gap — it makes knowledge vanish.

XAI's limit

Post-hoc explanations are reconstructions; false confidence is a real risk.

The principle

Transparency, like every principle, is contested and traded off — never free.

The black box is where transparency and responsibility collide.